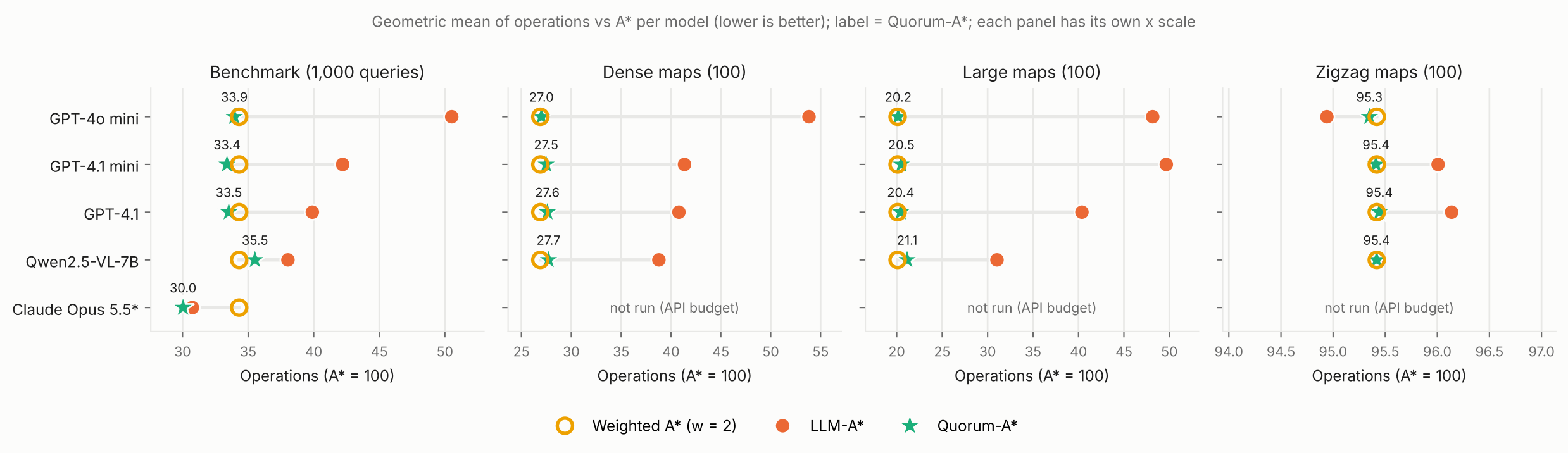
Geometric mean of operations vs A* per model (lower is better); label = Quorum-A*; each panel has its own x scale
Benchmark (1,000 queries)
Dense maps (100)
Large maps (100)
Zigzag maps (100)
GPT-4o mini
GPT-4.1 mini
GPT-4.1
Qwen2.5-VL-7B
Claude Opus 5.5*
33.9
33.4
33.5
35.5
30.0
27.0
27.5
27.6
27.7
20.2
20.5
20.4
21.1
95.3
95.4
95.4
95.4
not run (API budget)
not run (API budget)
not run (API budget)
30
35
40
45
50
25
30
35
40
45
50
55
20
25
30
35
40
45
50
94.0
94.5
95.0
95.5
96.0
96.5
97.0
Operations (A* = 100)
Operations (A* = 100)
Operations (A* = 100)
Operations (A* = 100)
Weighted A* (w = 2)
LLM-A*
Quorum-A*